Inter IIT Tech Meet 14.0 – VeQRA User Guide

This guide explains how to download, configure, and run the VeQRA project locally.
It can be accessed directly at https://unoverwhelmed-seclusively-jayna.ngrok-free.dev/

VeQRA consists of:,

Module	Tech
Frontend	React, Vite, Tailwind, GSAP, Framer-motion
Backend	Node.js, Express, PostgreSQL
Inference	Python + virtual environment
Database	PostgreSQL

Table 1: Tech Stacks

1. Clone the Repository

```
git clone "https://github.com/Team-76/ISRO_GI"
cd ISRO_GI
```

2. Frontend Setup

```
cd frontend
npm install
```

To run later:

```
npm run dev
```

Keep this terminal running.
The frontend will be available at: http://localhost:5173

3. Backend Setup

```
cd backend
npm install
mkdir uploads results
```

3.1. Configure Environment

Update /config.js with links appropriate to your deployment:

```
export const frontendLink = "http://localhost:5173";
export const backendLink = "http://localhost:5000";
export const modelLink = "http://localhost:8000";
```

4. Python Environment Setup

```
cd backend
python3 -m venv venv
```

Activate venv:

OS	Command
Windows	venv\Scripts\activate
Linux / macOS	source venv/bin/activate

Table 2: Venv

Install Python dependencies:

```
pip install -r requirements.txt
```

5. Set Up PostgreSQL

PostgreSQL is used to store users, chat sessions, and message history..

5.1. Install PostgreSQL

Download from https://www.postgresql.org/download
(If already installed, skip this step)

5.2. Open PostgreSQL Terminal

5.2.1. On Windows:

Search "SQL Shell (psql)" in Start Menu and open it.

5.2.2. On Linux / macOS:

Open regular terminal and run:

```
psql -U postgres
```

If the password is unknown:

```
sudo -u postgres psql
```

This logs into PostgreSQL as the postgres superuser without a password.

5.3. Create a new PostgreSQL user

```
CREATE USER veqra_user WITH PASSWORD 'your_password';
```

5.4. Create a database

```
CREATE DATABASE veqra_db OWNER veqra_user;
```

5.5. Grant Privileges (Extra Safety)

```
GRANT ALL PRIVILEGES ON DATABASE veqra_db TO veqra_user;
```

5.6. Connect to the New Database

```
\c veqra_db;
```

5.7. Update backend db.js

Inside backend/db.js, update:

```
user: "veqra_user",
host: "localhost",
password: "your_password",
database: "veqra_db",
port: 5432
```

6. Create Required Tables

Execute the SQL commands provided in postgres.txt (via psql or PgAdmin) to create:

Table	Purpose
users	Auth
chats	Stores uploaded images as sessions
messages	Stores LLM replies, grounding images, etc.

Table 3: PostgreSQL tables

Once the three tables are created, the backend is fully ready.

7. Run the Application

Open three terminals:

7.1. Terminal 1 — Frontend

```
cd frontend
npm run dev
```

The frontend runs at http://localhost:5173

7.2. Terminal 2 — Backend

```
cd frontend
npm run dev
```

The backend runs at http://localhost:5000

7.3. Terminal 3 — Python Inference

```
# activate virtual environment first
Windows: venv\Scripts\activate
Linux/macOS: source venv/bin/activate

python3 final_script.py
```

8. Completion

You can now:
- Upload an image
- Run captioning / grounding / LLM queries
- View saved chat history
- Switch dark/light mode

9. Using the Application

This section explains the workflow for interacting with the VeQRA web application after successful deployment.

9.1. Landing Page

Upon visiting the application URL, the user is directed to the landing page. The landing page contains the following elements:

- A brief description of the project and its purpose.
- A feature overview of the VeQRA system.
- A "Get Started" button located in the hero section.

Selecting the "Get Started" button redirects the user to the authentication page.

9.2. Authentication

The authentication interface provides two options:

Action	Required Fiels
Login	Email, Password
Sign Up	Name, Email, Password

Table 4: Auth

Upon successful login or account creation, the user is redirected to the main application interface.

9.3. Main Interface

The main interface consists of three primary components:

Component	Purpose
Image Area	Upload and view images
History Sidebar	Access previous chats and create new chat sessions
Query Panel	Submit queries related to the uploaded image

Table 5: Components

> Note: The Query Panel becomes visible only when an image is uploaded within the active chat session.

9.4. Uploading an Image

To begin an interaction:

1. Upload an image using the upload area in the main interface.
2. Once the image is uploaded:
   - The image becomes fixed in the display region.
   - A "Generate Caption" button appears below the uploaded image.
   - The image is automatically forwarded to the backend YOLO model for object detection.
   - The detected objects are displayed in the Query Panel as selectable tiles.

The user may trigger caption generation at any time by selecting the "Generate Caption" button.

9.5. Submitting Queries

Queries can be submitted using the Query Panel.
Selecting the Query Panel in the header opens the following input fields:

9.5.1. Query Type Selection

The user specifies the category of the question:

- Binary
- Numeric
- Semantic

9.5.2. Query Input Method

The user may submit the query using:
- Text input field, or
- Voice input (speech-to-text)

9.5.3. Language Selection

Multiple languages are supported and can be chosen from the dropdown menu.

9.5.4. Grounding Option (Object Selection)

All detected objects are listed as tiles.
The user may select the desired object for grounding before submitting the query.

9.6. Results and Display

- Responses to submitted queries appear as individual tiles to the right of the uploaded image.
- Responses are scrollable, while the uploaded image remains fixed.
- Newer responses appear at the top of the result list.

9.7. Managing Chats

- All chat sessions and associated queries are stored in the database.
- Users can view previous conversations by selecting entries from the History Sidebar.
- To start a new session, select the "New Chat" button located in the History Sidebar.

This opens a fresh chat interface where images and queries can be submitted independently of prior sessions.

9.8. Summary of User Workflow

- Visit landing page ⟶ select Get Started
- Log in / sign up
- Upload image in the main interface
- (Optional) Generate caption
- Submit query via Query Panel
- View response tiles beside the image
- Access history or begin a new chat at any time